UK-AIR Sensor Observation Service (Beta release)

Defra's UK-AIR Sensor Observation Service (SOS) provides a machine readable access point for air pollution measurements hosted on UK-AIR. The service supports UK obligations to European air quality e-Reporting initiative & Defra Open Data policy.

The UK-AIR SOS is based on the OGC Sensor Observation Service standard which has been extended to conform with INSPIRE download service specifications. The service provides an API type interface to UK-AIR for developers and machine to machine data transfer in open, machine readable data formats. It is not aimed at the general public nor traditional desktop air quality data users. We still recommend the resources on UK-AIR's data archive for desktop activities.

SOS Capabilities

The SOS aggregates readings from live, in-situ air quality monitoring stations and provides an interface to make sensors and sensor data archives accessible via an interoperable web based interface. The UK-AIR SOS is built on a 52°North SOS implementation which supports the OGC SOS versions 1.0.0, 2.0.0 and the European Air Quality e-Reporting data model (AQD version 1.0.0). A REST API is also supported.

The OGC SOS 2.0 specification has four extensions defined, three are include in the UK-AIR SOS; Core, Enhanced and Result Handling. The Transactional extensions is not implemented.

A full description of the UK-AIR SOS capabilities can be found at it's end point using a GetCapabilities request. A brief description is provided below.

Core Extension

- GetCapabilities, for requesting a self-description of the service
- GetObservation, for requesting the pure sensor data encoded in Observations & Measurements 2.0 (O&M)
- DescribeSensor for requesting information about a certain sensor, encoded in a Sensor Model Language 1.0.1 (SensorML 1.0.1) instance document.

Enhanced Extension

- GetFeatureOfInterest, for requesting the GML 3.2.1 encoded representation of the feature that is the target of the observation
- GetObservaitonById, for requesting the pure sensor data for a specific observation identifier

Transactional Extension

- Not implemented

Result Handling Extension

- InsertResultTemplate, not implemented.
- InsertResult, not implemented.
- GetResultTemplate, for getting the result structure and encoding for specific parameter constellations
- GetResult, for getting the raw data for specific parameter constellations

For more information about the SOS 2.0 specification, see the official OGC SOS 2.0 tutorial.

Supported Bindings

A binding describes how SOS clients and servers can communicate with each other (OGC#12-006). The following bindings are supported by the UK-AIR SOS.

- KVP - Key Value Pair encoding of requests via HTTP GET.
- SOAP - SOAP based encoding of all extensions via HTTP POST.
- POX - XML based encoding of all extensions via HTTP POST (without SOAP wrapping).
- REST - RESTful binding for all useful operations of resources being available in the SOS. Download the documentation (PDF 880 KB).
- JSON - JSON based encoding of all extensions via HTTP POST.
- EXI - Efficient XML Interchange (EXI) Format 1.0 based encoding of all extensions via HTTP POST and POX encoded request. SOAP requests are currently not supported via the EXI binding.

Pollutant coverage

A list of the pollutants included in the UK-AIR SOS is provided on the SOS Pollutants page.

Querying the SOS

Some examples of how the SOS can be queried are provided in two common bindings (KVP and JSON) which are available to download (PDF 273 KB). Download documentation for the REST API.

Data Archive

- $PM_{2.5}$ Targets (PERT and AMCT)
- Methodology for calculating progress towards the $PM_{2.5}$ targets
- Data Selector
- Annual and Exceedance Statistics
- Data availability
- DAQI regional data
- Air quality compliance data
- UK Air Quality Data Catalogue
- UK Particle Monitoring Programme
- Marylebone Road, Non-Automatic Data
- PAH data
- TOMPs data
- Metals data
- Comparing data
- UK Ambient Air Quality Interactive Map
- Modelled air quality data
- Openair - data analysis tool
- Background Maps
- Stratospheric Ozone Data
- UV Radiation Data
- UV Index Graphs
- WMS Services
- Spatial Objects
- Atom Download Services
- **Sensor Observation Service**
- Population Density of UK
- Critical Loads
- Licences of air quality mapping data

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